

Transformations, Groups and Geometry

UNSW MATH3511 25T3

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Contents

0	Revision	2
1	Triangle Geometry	2
1.1	Signed Lengths and Orientation	2
1.2	Intersections	2
1.3	Collinearity	3
1.4	Angle Bisectors	3
1.5	Area Formulas	4
1.6	Circles Associated to Triangles	5

0 Revision

Here is a list of concepts you should know from high school.

- Vertically opposite, alternating, corresponding and co-interior angles.
- Interior angles of a triangle sum to π ; exterior angle equals sum of opposite interior angles.
- Congruent and similar triangles.
- Isosceles and equilateral triangles.
- Sine rule: $\frac{a}{\sin \alpha} = \frac{b}{\sin \beta} = \frac{c}{\sin \gamma}$.
- Cosine rule: $a^2 = b^2 + c^2 - 2bc \cos \alpha$. Pythagoras' theorem if $\alpha = \frac{\pi}{2}$.
- Three non collinear points defines a unique circle.
- A quadrilateral is cyclic if its vertices lie on a circle.

1 Triangle Geometry

1.1 Signed Lengths and Orientation

A line through two points A and B is denoted AB , typically *in that direction*. For a point P on the line AB , if P lies between A and B then the ratio $\frac{AP}{PB} > 0$. But if P is outside A and B then $\frac{AP}{PB} < 0$. If P and A are the same point, then the ratio is zero; if P and B are the same, then the ratio is ∞ .

An **altitude** of a triangle is a line segment through a given vertex and perpendicular to the line containing the side opposite the vertex.

In a triangle, the orientation is called **positive** if the vertices are taken anti-clockwise in the order given. For an oriented triangle, we take directed lengths to match the orientation: AB, BC and CA are positive in $\triangle ABC$. We extend this convention to area and angle:

- The area of a triangle is positive if the triangle is positively oriented; otherwise it is negative.
- Positive angles go anticlockwise. If edges are signed, we need to account for that. We take $\angle ABC$ as positive if the sense runs anticlockwise from BA to BC when both are the same sign; if they are opposite signs, we change the sign of the angle to compensate.

We denote the sign of the area as $\triangle ABC = \frac{1}{2}|AB| \cdot |BC| \cdot \sin \angle ABC$.

1.2 Intersections

A **centroid** is the intersection of the medians while a **orthocetnre** is the intersection of altitudes. It can be shown that the medians are concurrent. The medians and altitudes are examples of **cevians**.

Definition. (Cevian) In any triangle a line segment joining a vertex to a non-vertex point on the opposite side, possibly extended, is called a cevian.

Theorem. (Ceva's Theorem (1678)) In $\triangle ABC$ the cevians AD , BE and CF (where D is on BC , E is on AC and F is on AB) are concurrent if and only if

$$\frac{AF}{FB} \cdot \frac{BD}{DC} \cdot \frac{CE}{EA} = 1.$$

Corollary. In $\triangle ABC$ the cevians AD , BE and CF are concurrent if and only if

$$\frac{\sin ACF}{\sin FCB} \cdot \frac{\sin BAD}{\sin DAC} \cdot \frac{\sin CBE}{\sin EBA} = 1,$$

taking proper account of orientations of the angles.

1.3 Collinearity

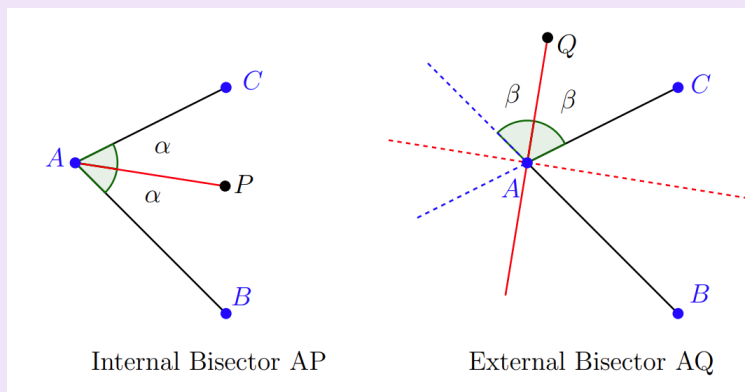
Theorem. (Menelaus' Theorem) In $\triangle ABC$ let D , E and F be on side BC , CA , AB (or their extensions) respectively. Then D , E and F are collinear if and only if

$$\frac{AF}{FB} \cdot \frac{BD}{DC} \cdot \frac{CE}{EA} = -1.$$

Theorem. (Pappus' Theorem) Suppose points A, C, E lie on one line and B, D, F on a second line, where AB , CD and EF form a triangle. Let $L = AB \cap DE$, $M = CD \cap FA$ and $N = EF \cap BC$. Then L , M and N are collinear.

1.4 Angle Bisectors

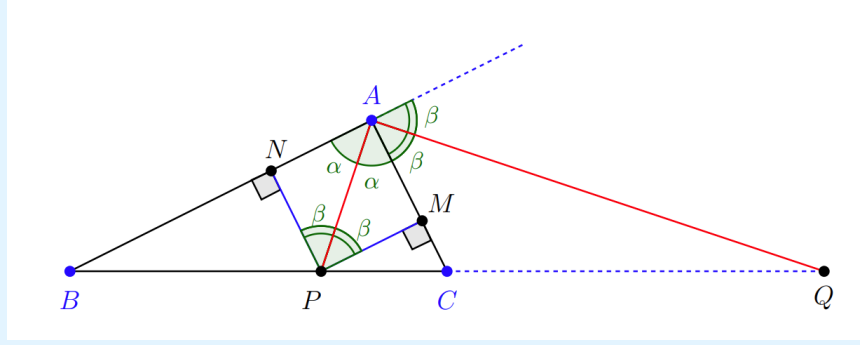
Definition. (Bisectors) In the image below, the line AP in the left hand diagram is the **internal bisector** of $\angle BAC$: $\angle BAP = \frac{1}{2}\angle BAC$ and the line AQ in the right hand diagram is the **external bisector** of $\angle BAC$: $\angle CAQ = \frac{1}{2}(\pi - \angle BAC)$, that is it bisects angle between AC and the extension of the line AB .



Theorem. (Angle Bisector Theorem) Consider $\triangle ABC$, and its angle bisector as below.

a) If the interior bisector of $\angle BAC$ meets BC at P then $\frac{BP}{PC} = \frac{AB}{CA}$.

b) If the exterior bisector of $\angle BAC$ meets BC at Q then $\frac{BQ}{QC} = -\frac{AB}{CA}$.



Theorem. (Incentre Theorem) The internal bisectors of a triangle are concurrent.

The point of concurrence is called the **incentre**. This point is equidistance from the three sides, so a circle centred here and tangent to the edges is the largest circle that can fit inside the triangle. This circle is called the **incircle**.

Theorem. (Excentre Theorem) The external bisectors of any two angles of a triangle are concurrent with the internal bisector of the third angle.

The points of concurrence (there are three of them) are called the **excentres**. Again, each excentre is the same distance from the three sides, and so circles centres at the excentres with that distance as radius (the **excircles**) are tangent to the sides of the triangle, one internally, two externally.

Theorem. In $\triangle ABC$, let the internal bisectors at angles A, B and C meet the opposite sides at P, Q and R respectively and the external bisectors meet the opposite sides at P', Q' and R' respectively. Then $\{P, Q, R'\}$, $\{P, Q', R\}$ and $\{P', Q, R\}$ are triples of collinear points.

1.5 Area Formulas

Let r be the radius of the incircle. Then

$$\triangle ABC = \frac{1}{2}r(a + b + c) = sr,$$

where $s = \frac{1}{2}(a + b + c)$ is the semiperimeter.

We can get similar results from the excircles. Let r_a be the radius of the excircle centred at J_a . Then

$$\triangle BCJ_a = \frac{1}{2}r_a \cdot a, \quad \triangle ABJ_a = \frac{1}{2}r_a \cdot c, \quad \triangle ACJ_a = \frac{1}{2}r_a \cdot b,$$

since r_a is the distance to each edge. Thus,

$$\triangle ABC = \frac{1}{2}r_a(b - a + c) = (s - a)r_a.$$

Then similarly, we have

$$\triangle ABC = (s - b)r_b = (s - c)r_c.$$

After some trigonometry and rearranging to solve for r , we get

$$r = \sqrt{\frac{(s-a)(s-b)(s-c)}{s}}.$$

Then from the previous expression we get **Heron's formula**:

$$\Delta ABC = \sqrt{s(s-a)(s-b)(s-c)}.$$

Then we get the following results,

$$r_a = \sqrt{\frac{s(s-b)(s-c)}{s-a}} \quad \text{and} \quad \frac{1}{r} = \frac{1}{r_a} + \frac{1}{r_b} + \frac{1}{r_c}.$$

1.6 Circles Associated to Triangles

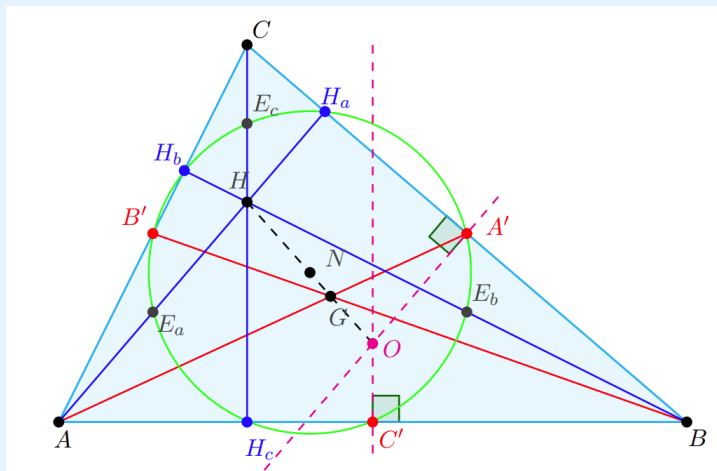
The **circumcircle** goes through the three vertices of a triangle and has centre the **circumcentre**, the intersection of the three perpendicular bisectors of the sides.

Suppose for a circle $\triangle ABC$ with circumcentre O . Let $OA = OB = OC = R$ be the **circumradius**. Then $\angle AOB = 2\angle C$ and so $\angle C = \angle FOB$. Thus $R \sin C = FB = \frac{c}{2}$. Then similar results for the other sides gives the **extended sine rule**:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R.$$

Lemma. The circumcentre of a triangle is the orthocentre of its median triangle.

Theorem. (Nine Point Circle) For $\triangle ABC$, let A', B' and C' be the midpoints of the three sides and H_a, H_b and H_c be the feet of the altitudes. Let H be the orthocentre and E_a, E_b and E_c be the midpoints of the lines from H to the three vertices. Also let O be the circumcentre and R the circumradius. Then if N is the midpoint of the line from H to O , the nine points $A', B', C', H_a, H_b, H_c, E_a, E_b$ and E_c lie on the circle centre N of radius $\frac{R}{2}$.



Theorem. (Feuerbach's Theorem) The nine-point circle is tangent to the incircle and all three excircles.

The point where the incircle and the nine-point circle meet is called the **Feuerbach point**. It is collinear with the incentre and the nine-point centre.

Theorem. (The Euler Line) In any triangle, the orthocentre H , nine-point centre N , circumcentre O and centroid G are collinear. Furthermore N is the midpoint OH and $OG : GH = 1 : 2$.

The line these points lie on is called the **Euler Line**.

Theorem. (Simson Line) Let P be any point on the circumcircle of $\triangle ABC$. Then the feet of the perpendiculars from P to each side of $\triangle ABC$ are collinear.

Proposition. The three Euler points uniquely define a triangle.